



EXAM PAPERS PRACTICE

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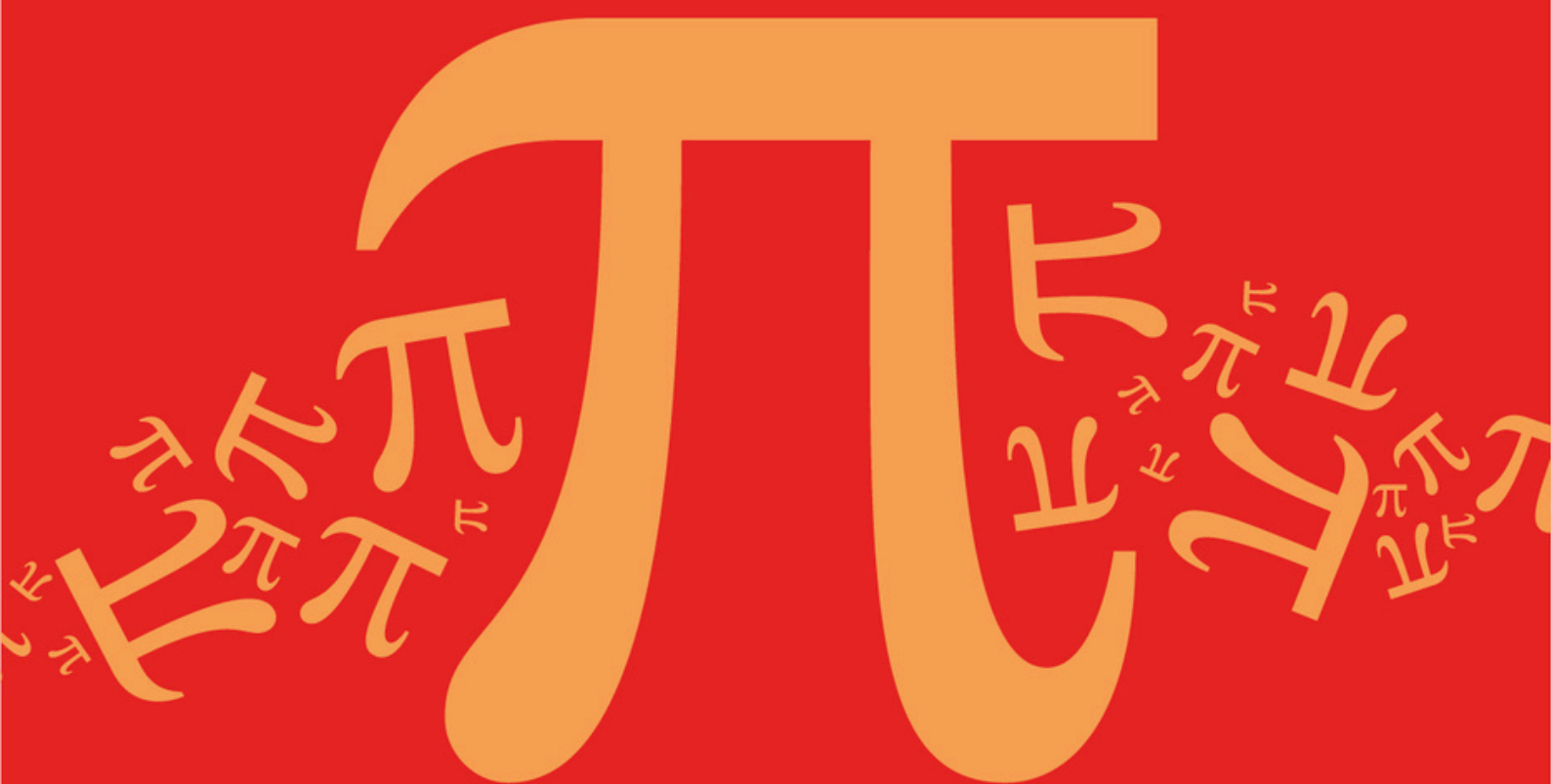
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AQA A Level Further Mathematics (7367)



Topic

DA: Graphs

Sub topic

DA7: Graph Isomorphisms

Topic Questions

AQA A Level Further Mathematics (7367)

Question Paper

Topic **DA: Graphs**
Sub topic **DA7: Graph Isomorphisms**

EXAM PAPERS PRACTICE

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**To be used by all students and teachers for
AQA A Level Further Mathematics (7367)
for exams from 2025 onwards.**

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Name : _____

Date : _____

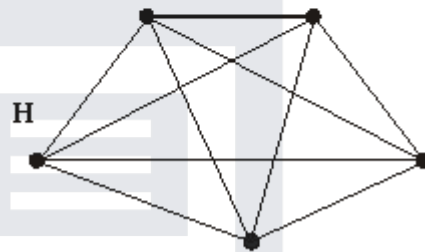
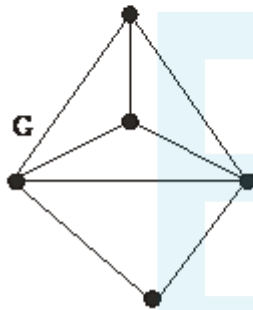
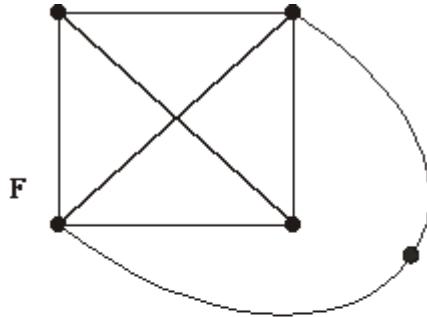
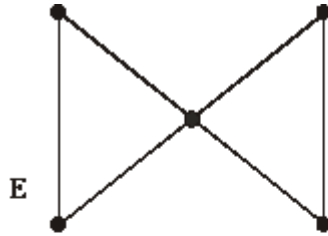
Time : _____

Total Marks Available : _____

Total Marks Received : _____

Q1.

The four graphs **E**, **F**, **G** and **H** are illustrated below. They each have five vertices.

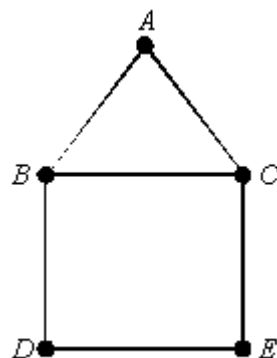


- (a) State which, if any, of the four graphs are Eulerian. (2)
- (b) State which, if any, of the four graphs are Hamiltonian. (2)
- (c) State which, if any, of the four graphs are planar. (2)
- (d) State which pairs, if any, of the four graphs are isomorphic. (2)

(Total 8 marks)

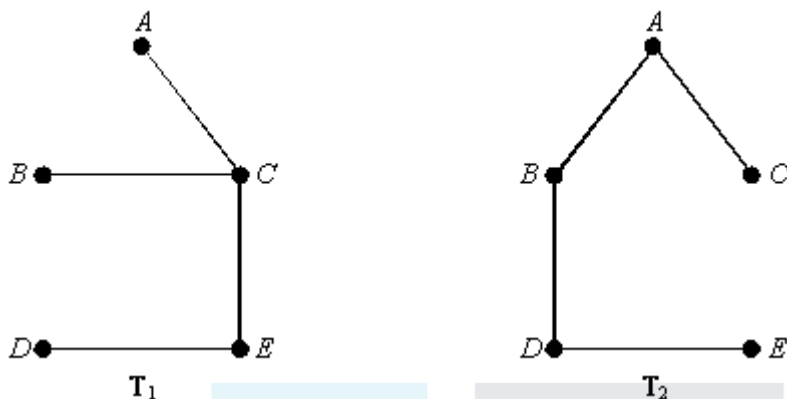
Q2.

The graph **G** is illustrated:



Graph G

Two of G 's spanning trees (or minimum connectors, with all the edges of equal weight) T_1 , and T_2 are shown below. In each of these trees the degree of the vertex A equals the degree of the vertex D .

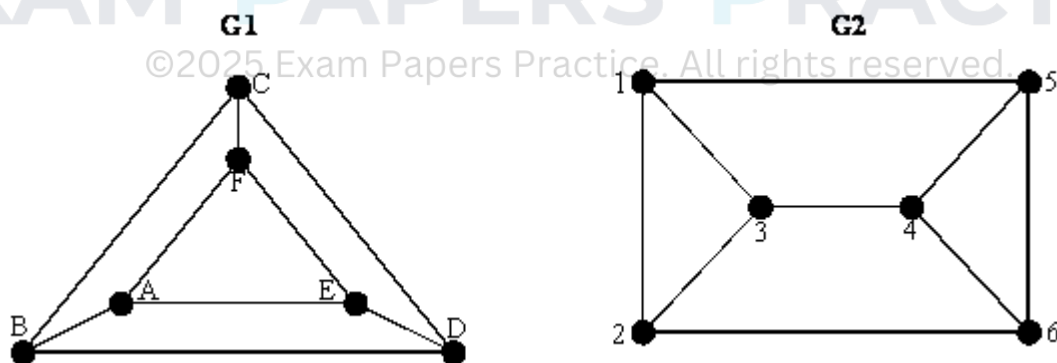


- (a) Draw **three** other spanning trees of G in which the degree of vertex A equals the degree of vertex D . (6)
- (b) For each of the trees which you found in part (a), state whether it is isomorphic to T_1 , or to T_2 or to neither. (3)

(Total 9 marks)

Q3.

- (a) Show that the following two graphs are isomorphic.



- (b) Draw a simple connected graph on six vertices, each of degree 3, which is not isomorphic to the graphs in part (a). (3)

(2)
(Total 5 marks)

Q4.

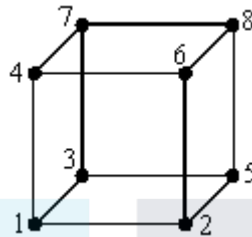
A pumping room contains three large electrically powered pumps, pump A, pump B and pump C. They are all currently pumping. Any pump can be switched off if it is on (or on if it is off), but only one at a time may be switched to avoid damaging power surges.

- (a) Draw a graph in which the vertices represent the possible states (on or off) of each

pump, and in which the edges connect pairs of vertices which differ by just one switching. The state when all three are on is represented by the vertex (A, B, C) , and this vertex is connected by an edge to the vertex $(\sim A, B, C)$, and to some other vertices.

(3)

- (b) Show that your graph in part (a) is isomorphic to the graph shown below.



(2)

- (c) Give the number of paths with three edges connecting (A, B, C) to $(\sim A, \sim B, \sim C)$.

(2)

- (d) Give the number of paths which have 5 edges, none repeated, connecting (A, B, C) to $(\sim A, \sim B, \sim C)$.

(2)

(Total 9 marks)

Q5.

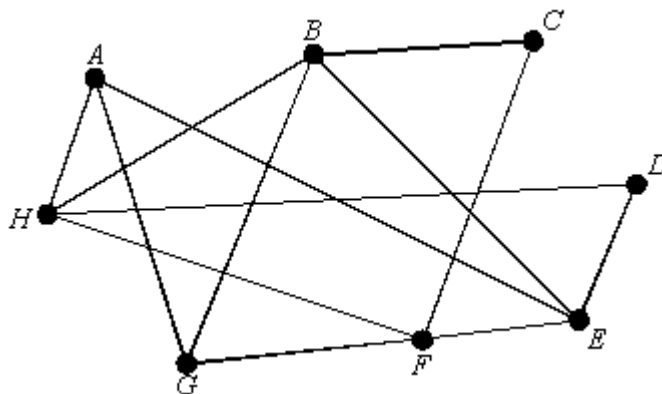
- (a) Draw a sketch of the graph K_5 .

(1)

- (b) Draw a sketch of the graph $K_{3,3}$.

(1)

The graph G is shown:



- (c) Draw a subgraph of G which is isomorphic either to K_5 or to $K_{3,3}$.

(2)

(Total 4 marks)